COURSE DESCRIPTION FORM: DS-5007: Natural Language Processing

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad Campus

MS-DS: **Spring-2026**

PROGRAM TO BE EVALUATED

Course Description

Course Code	DS-5007
Course Title	Natural Language Processing
Credit Hours	3
Course Instructors	Dr. Zohair Ahmed
Grading Policy	Absolute grading
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. Student who misses an assessment item (other than sessional / final exam) is awarded zero marks in that assessment item i.e. late submission will not be accepted. For missed sessional/ final exam, exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.
Course Plagiarism Policy	Plagiarism in project or sessional/ final exam will result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.
Prerequisites by Course(s) or Topics	Preferred but not necessary: Data Learning
Assessment Instruments with Weights (homeworks, quizzes, sessional exams, final exam, assignments, etc.)	Assessment with the weight. **Assessment Type** – **Weight** Assignments – 10% Quizzes – 10% Sessional Exams (2) – 20% Project – 20% Finals – 40%
Course Coordinator	Dr. Zohair Ahmed
URL (if any)	
Course Catalog Description	This course presents a comprehensive study of both traditional and modern approaches to Natural Language Processing. It begins with fundamental text preprocessing techniques and introduces statistical methods with an emphasis on probabilistic and Bayesian modeling. Particular attention is given to language modeling, its challenges, and class-conditional formulations.

	The course then transitions to modern Natural Language Processing paradigms based on deep learning, starting with distributed word representations that address key limitations of classical language models. Advanced topics include neural sequence modeling and transformer-based architectures, providing students with a solid foundation in contemporary Natural Language Processing systems.
Textbook	Daniel Jurafsky and James H. Martin.. **Speech and Language Processing: An Introduction to Natural Language Processing**. Third Edition (or the latest). Prentice Hall
Reference Material	Steven Bird, Ewan Klein and Edward Loper.2019. Analyzing Text with the Natural Language Toolkit. Natural Language Processing with Python. O'Reilly. Proceedings of ACL, EMNLP, NAACL, TACL
Course Goals	

A. Course Learning Outcomes (CLOs)

After course completion, the students shall be able to:

1. Analyze and evaluate traditional and statistical Natural Language Processing techniques used for language modeling and text analysis.
2. Assess advanced neural and transformer-based architectures for Natural Language Processing, including large language models, in terms of design principles and limitations.
3. Evaluate advanced Natural Language Processing system architectures, including retrieval-augmented and multimodal approaches, with respect to performance, robustness, and ethical considerations.
4. Formulate and present a research-oriented solution or system design in Natural Language Processing by synthesizing recent literature and experimental findings.

B. Program Learning Outcomes (PLOs)

1. To equip students to transform data into actionable insights to make complex business decisions.
2. To enable students, understand and analyze a problem and arrive at computable solutions.
3. To expose students to the set of technologies that match those solutions.
4. To gain Hands – on experience on data- centric tools for statistical analysis. Visualizations. And big data applications at the same rigorous scale as in a practical data science project.
5. To understand the implications of handling data in terms of data security and business ethics.
6. Students shall have the ability to make effective oral and written presentations on technical topic

C. Mapping of CLOs to PLOs

(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)

		PLOs									
		1	2	3	4	5	6				
	1		✔	✔			✔				

CLOs	2			✔		✔					
	3	✔			✔						
	4		✔				✔				

Topics covered in the course
(assume 15-week instruction and 3 contact hours per week)

Topics to be covered:

List of Topics	No. of Weeks	Contact Hours	CLO(s)
Natural Language Processing foundations: history, applications, tasks, datasets, and annotation.	**1**	**3**	
Text processing and normalization: Unicode, cleaning, regular expressions, tokenization, and sentence segmentation.	**1**	**3**	
Classical text representations and models: Bag-of-Words, TF-IDF, Naïve Bayes, and Logistic Regression.	**1**	**3**	
Evaluation methods for NLP: confusion matrix, precision, recall, F1-score, cross-validation, and basic error analysis.	**1**	**3**	
Language modeling fundamentals: N-gram models, perplexity, and smoothing concepts.	**1**	**3**	
Vector semantics and word embeddings: co-occurrence, PMI, Word2Vec, GloVe, and limitations of static embeddings.	**1**	**3**	
Sequence labeling: POS tagging, Named Entity Recognition, BIO tagging schemes, and entity-level evaluation.	**1**	**3**	
Neural NLP foundations: batching, padding, masking, loss functions, regularization, and training pipelines in PyTorch.	**1**	**3**	

	Recurrent neural networks in NLP: RNN, LSTM, GRU architectures and their use in text classification and NER.	**2**	**3**	
	Transformer architectures: self-attention, multi-head attention, positional encodings, encoder/decoder roles, and HuggingFace ecosystem.	**1**	**3**	
	BERT-style models: masked language modeling, contextual embeddings, and fine-tuning for downstream NLP tasks.	**1**	**3**	
	GPT-style models: decoder-only language models, text generation strategies, and prompting techniques.	**1**	**3**	
	LLM adaptation methods: prompt engineering, parameter-efficient fine-tuning (PEFT), and Low-Rank Adaptation (LoRA).	**1**	**3**	
	Retrieval-Augmented Generation (RAG): document chunking, embedding-based retrieval, vector databases, grounding, and hallucination mitigation.	**1**	**3**	
	NLP system evaluation: benchmarking, bias and fairness analysis, robustness testing, and advanced error analysis.	**1**	**3**	
	Total	**16**	**48**	
Programming Language for Assignments	Programming assignments will be implemented in Python using widely adopted Natural Language Processing and deep learning libraries, including NLTK, spaCy, TensorFlow, PyTorch, and Keras, to support model development, training, and evaluation.			

Class Time Spent (in percentage)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	55	20	20	5
Oral and Written Communications	Every student is required to submit at least___5____written reports of typically____5_____ pages each and to make___1____oral presentation of typically_____10___minutes' duration.			